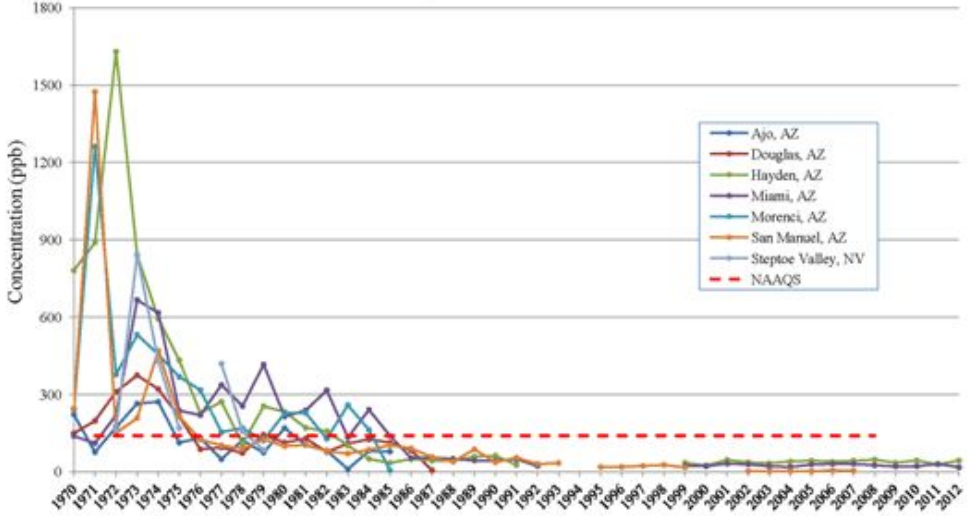
Stable Isotopes: Lichens as an Emission Proxy

Orion Hoch

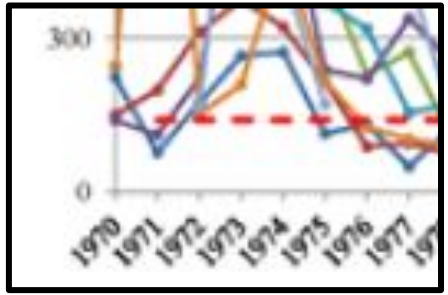
Research Question:

With stable isotopes, can we use lichens as a proxy for measuring atmospheric pollution historically?



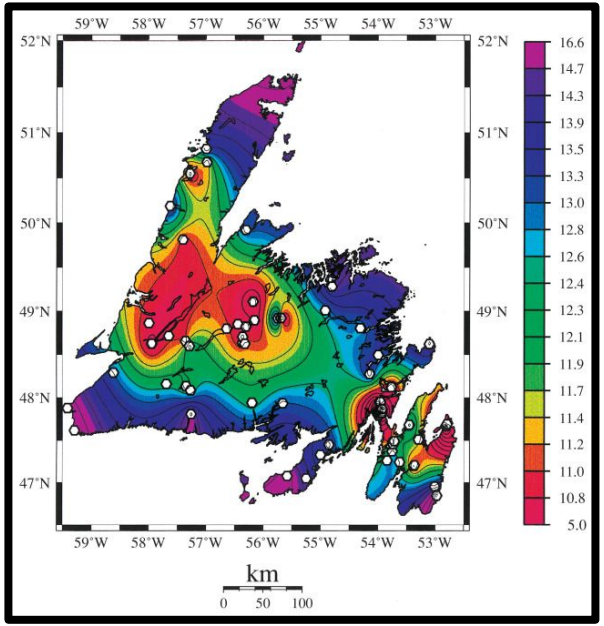
*As designated under the 1971 sufur dioxide (SO_2) primary national ambient air quality standard (NAAQS).

Source: US EPA's Air Quality Systems (AQS) database (August 31, 2012 and June 25, 2013, last updated July 15, 2013). The 1971 NAAQS for 24-hour SO_2 is 140 parts per billion (ppb), not to be exceeded more than once in a calendar year. The form of the design value for primary 24-hour SO_2 is the 2nd-highest average value, based on hour data that are at least 75% complete for each calendar quarter. Not all design values are valid due to incomplete data. The primary annual and 24-hour SO_2 standards were revoked in 2010. No monitoring data is available for the nonattainment areas in Guam.



Expectations/Prev. Studies

- Lichens have been proven to not significantly shift delta values (fractionate) of specific elements
- Previous studies have focused on spatial resolution but not much has been done on the historical record
- Attempting to gain a better understanding of how emission sources have changed relative to our policies

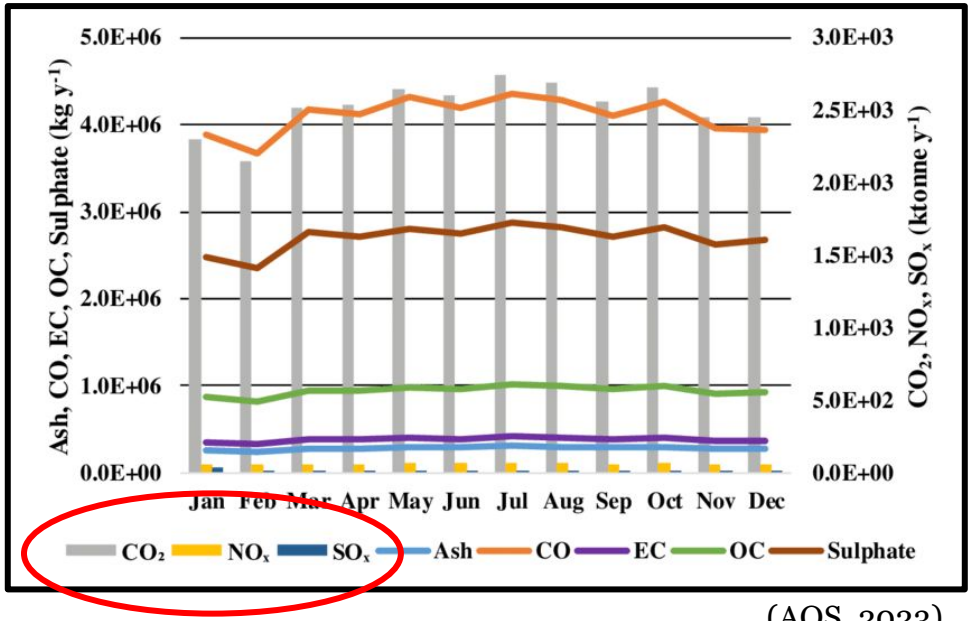


(Wadleigh. 1999)

Emission Partitioning

Nitrogen, Sulphur, and Carbon make up 3 of the largest atmospheric pollutants in NO_x, SO_x, and CO_2

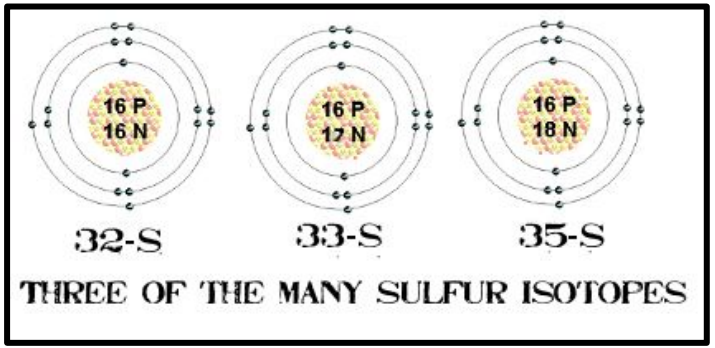
All have distinct stable isotopes when analyzing their sources

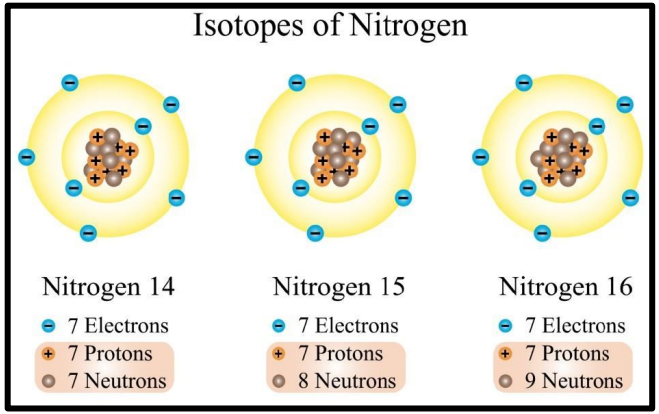


(AQS. 2023)

Stable Isotopes Review

- Same element, different number of neutrons
- Different mass
- We can use ^{34}S, ^{15}N and ^{13}C to measure pollutants!



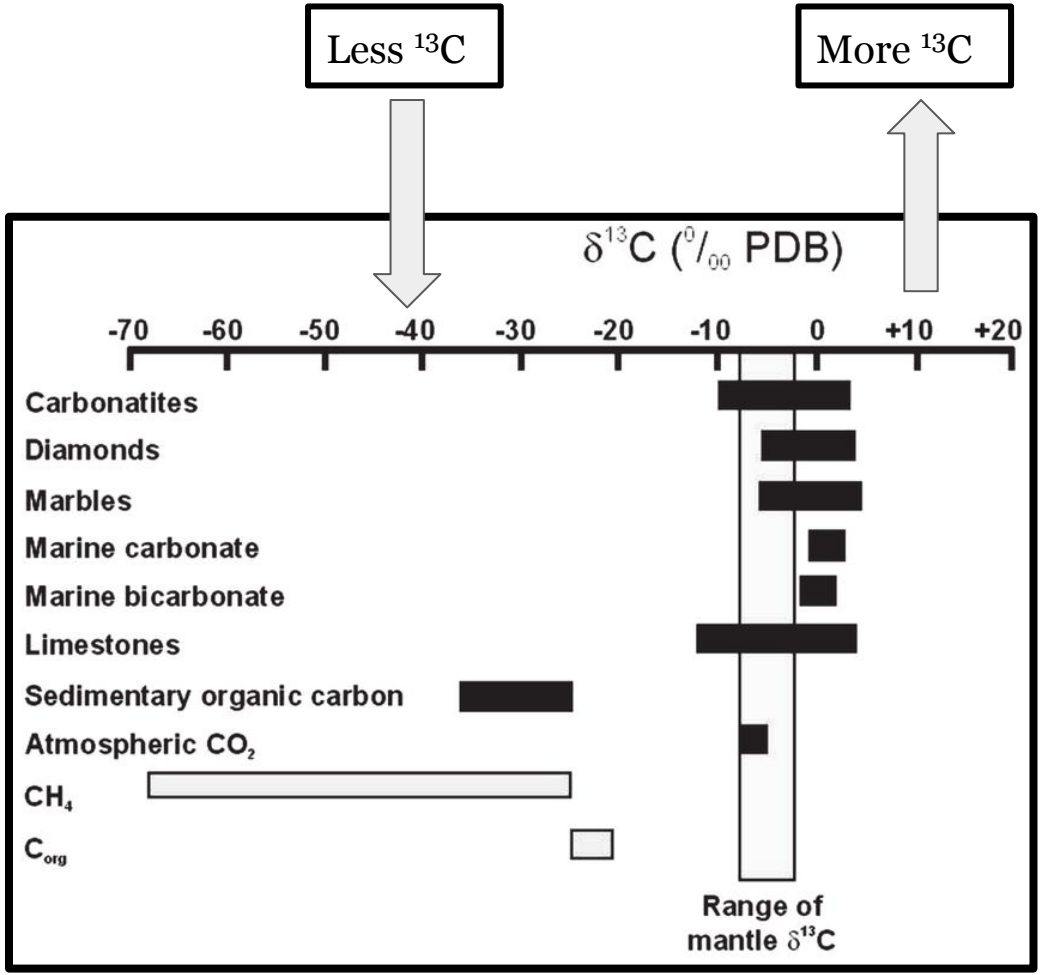


δ Delta Values

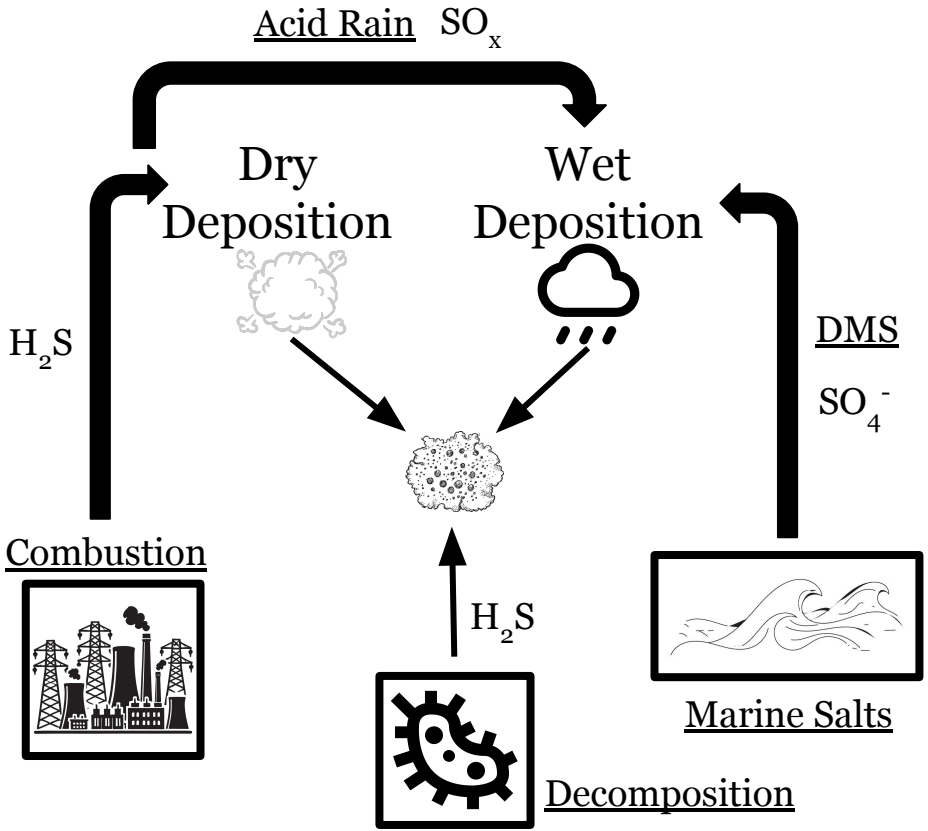
Positive Values represent a higher ratio and hence isotopically **HEAVIER** molecule

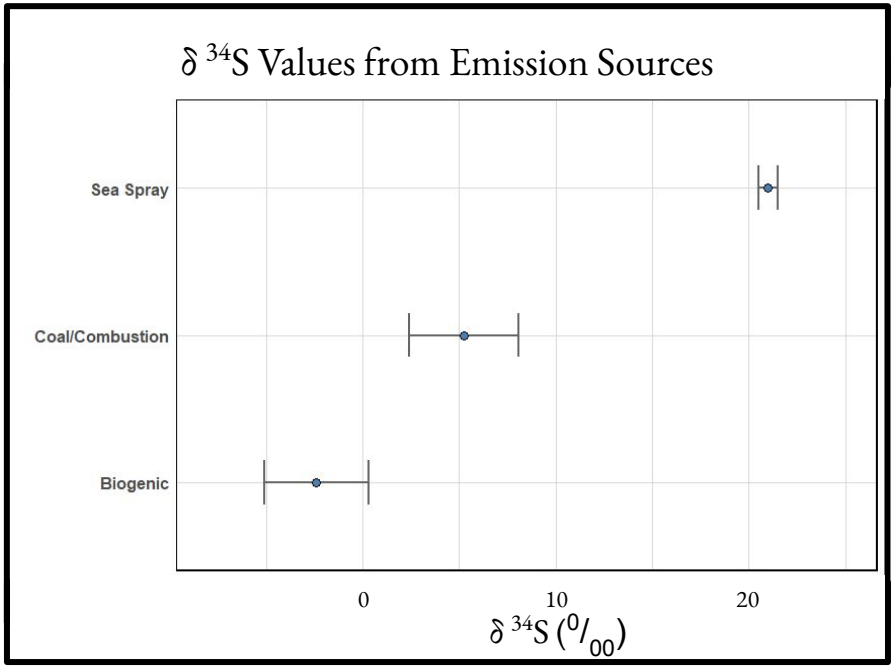
Negative Values present a lower ratio and **LIGHTER** molecule

Zero is just the scientific standard for that element and usually chosen because of distinct signature

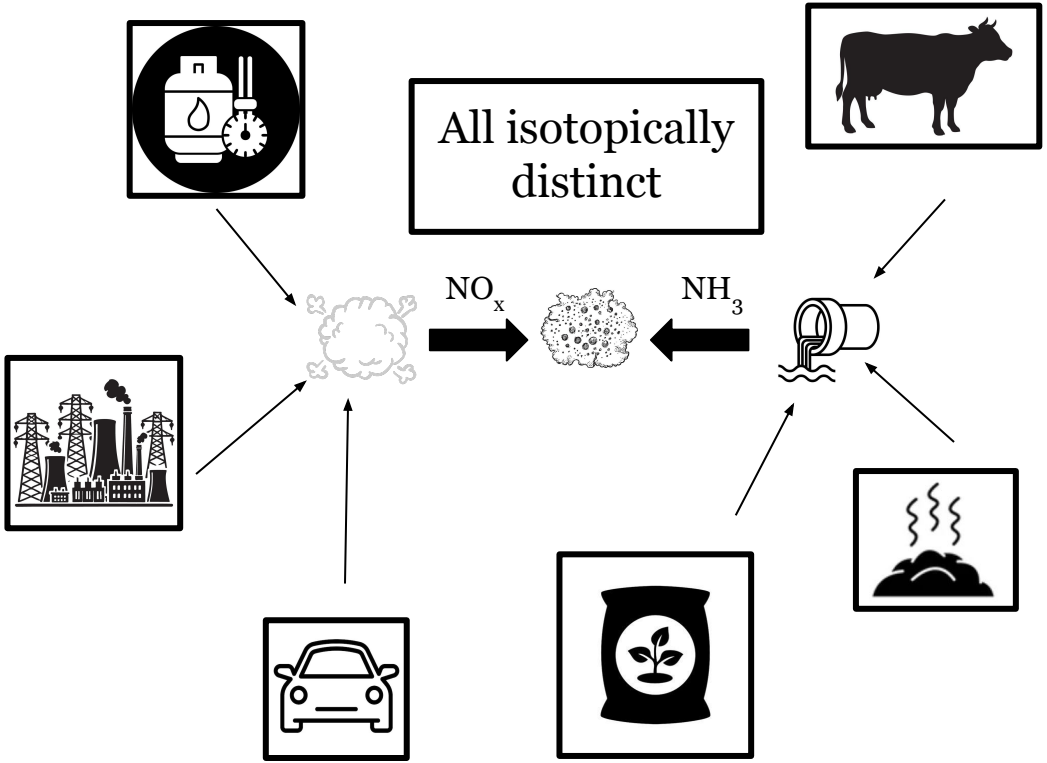


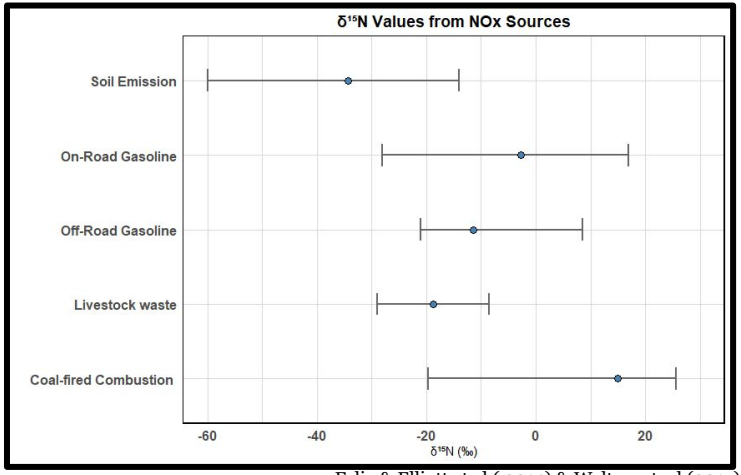
Sulphur Pathways



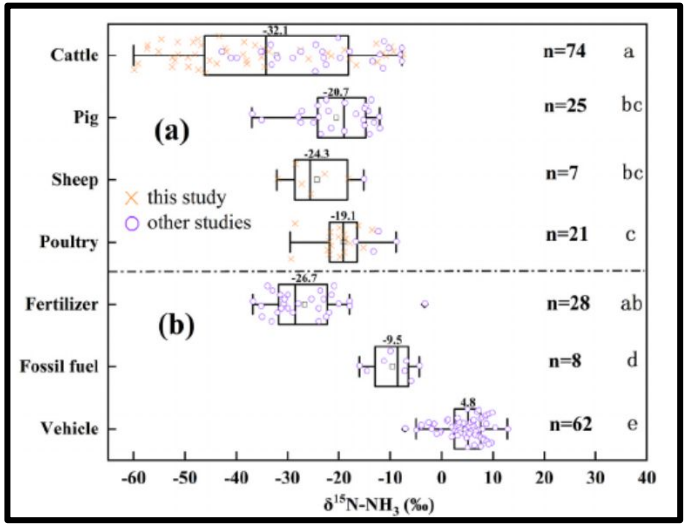


Nitrogen Pathways





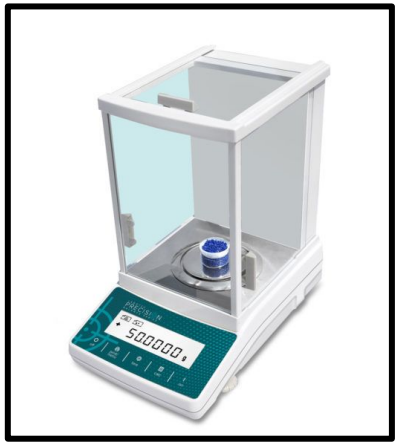
Felix & Elliott et al.(2014) & Walters et. al (2015)

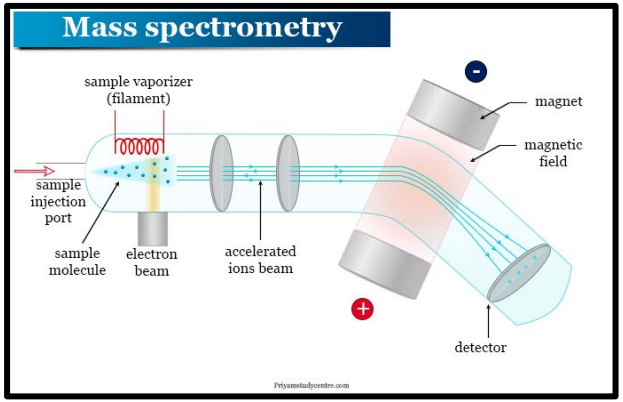


Liu et al. (2025)

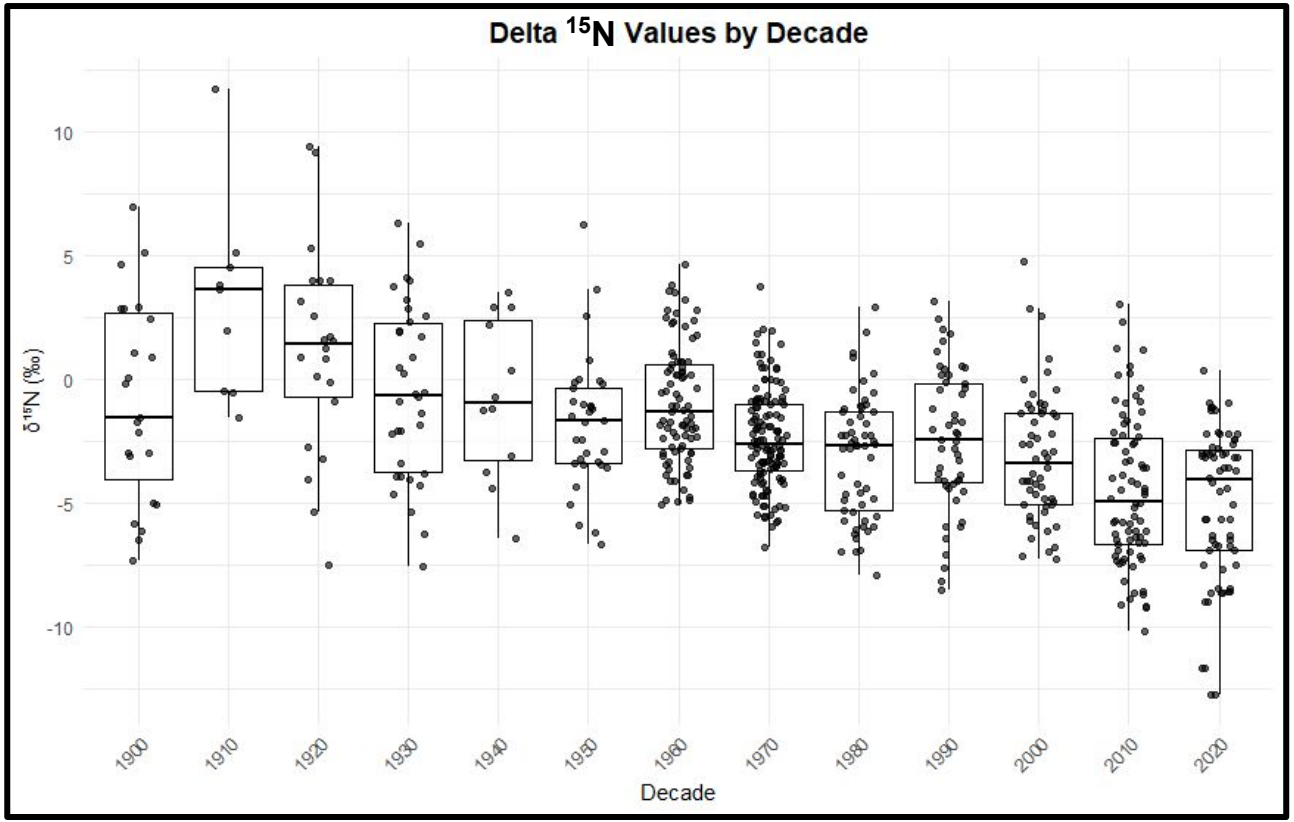
Methodology

- Collect sample from anthropogenic collection
- Prepared by Hannah Monti with samples from NY Botanical Garden and NY State Museum
- Crush and encapsulate lichen samples
- Isotope Ratio Mass Spectrometer



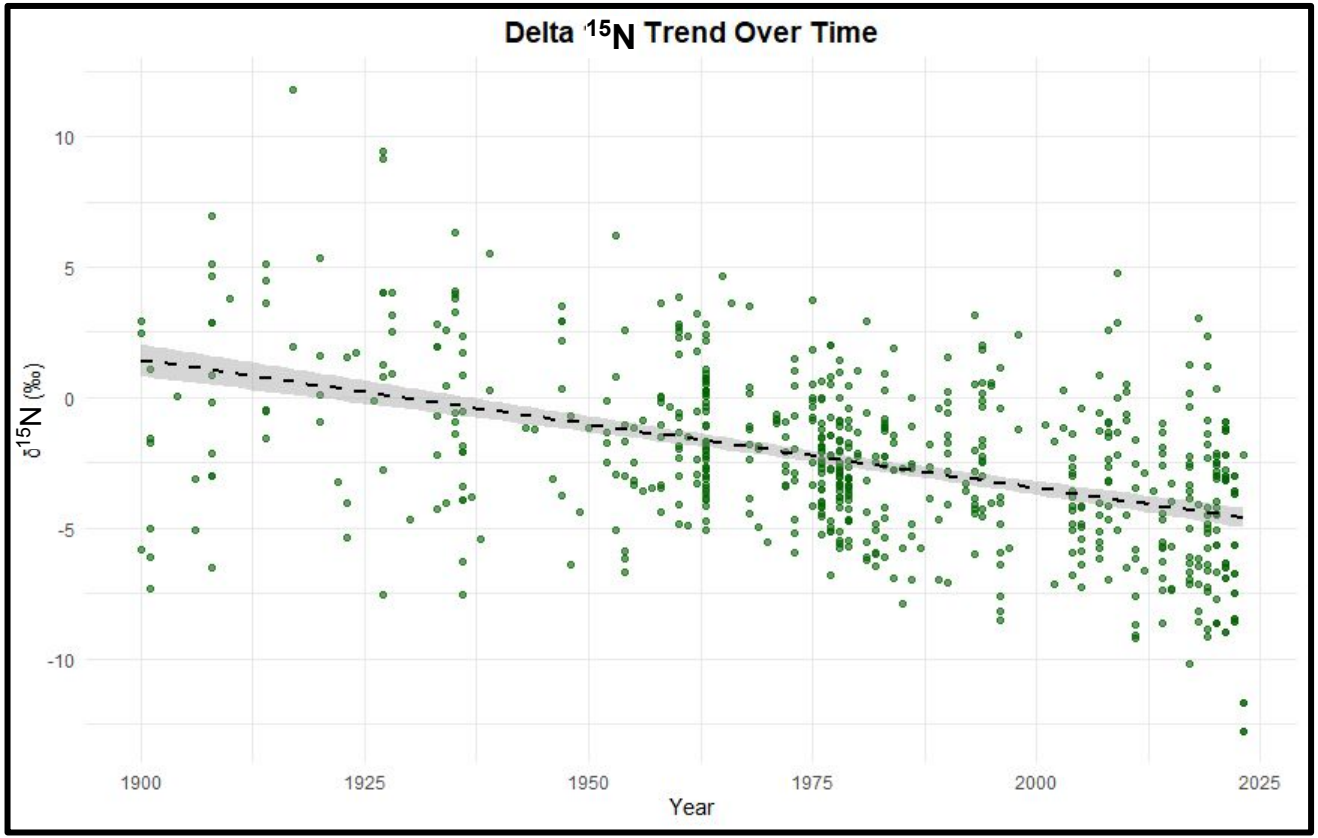


Results - Nitrogen



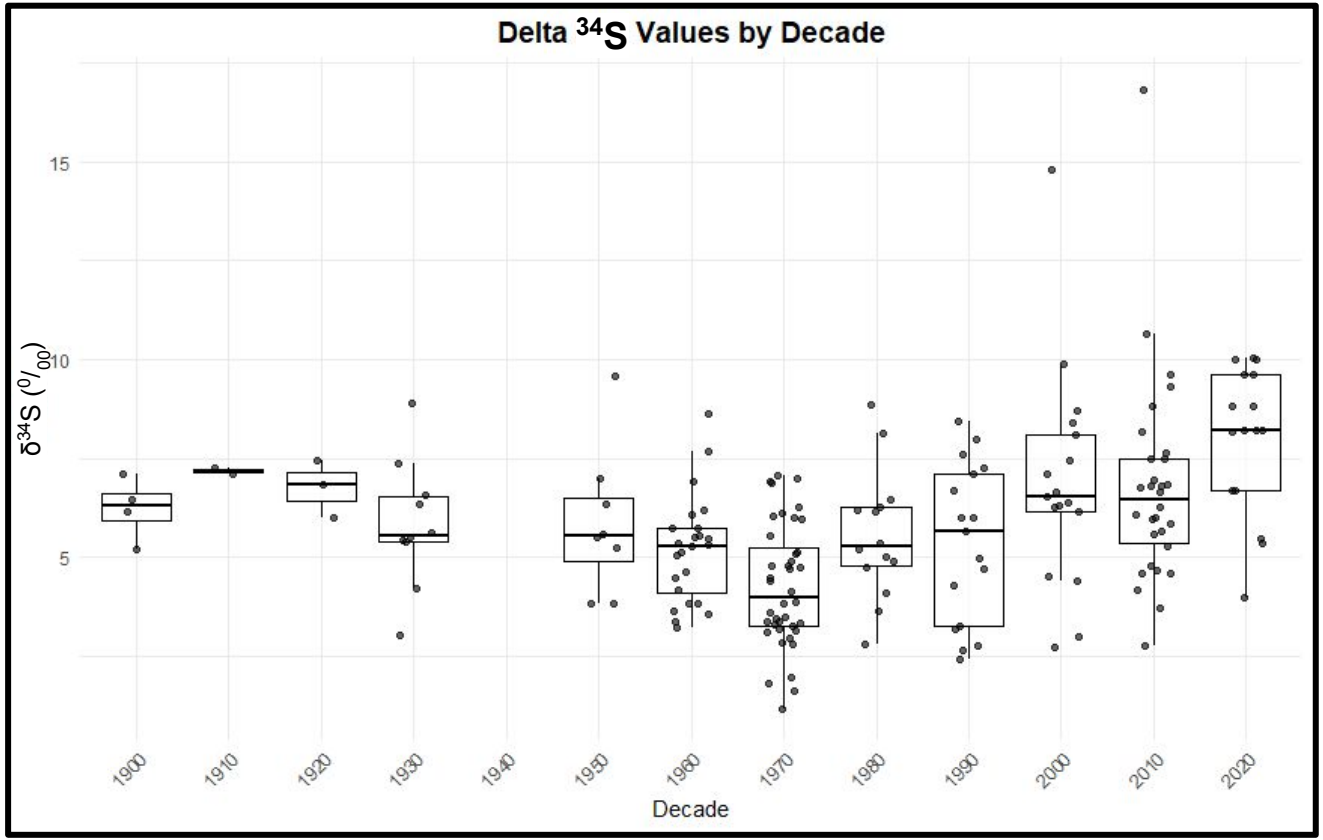
- Early sources of NO_x contamination
  - Coal (+15 NO_x)
  - Higher avg **δ**
- N sources shifted in late 20th century
  - Fossil Fuels (-9.5 NH_x, -11.5 NO_x)
  - Lower avg **δ**

Results - Nitrogen



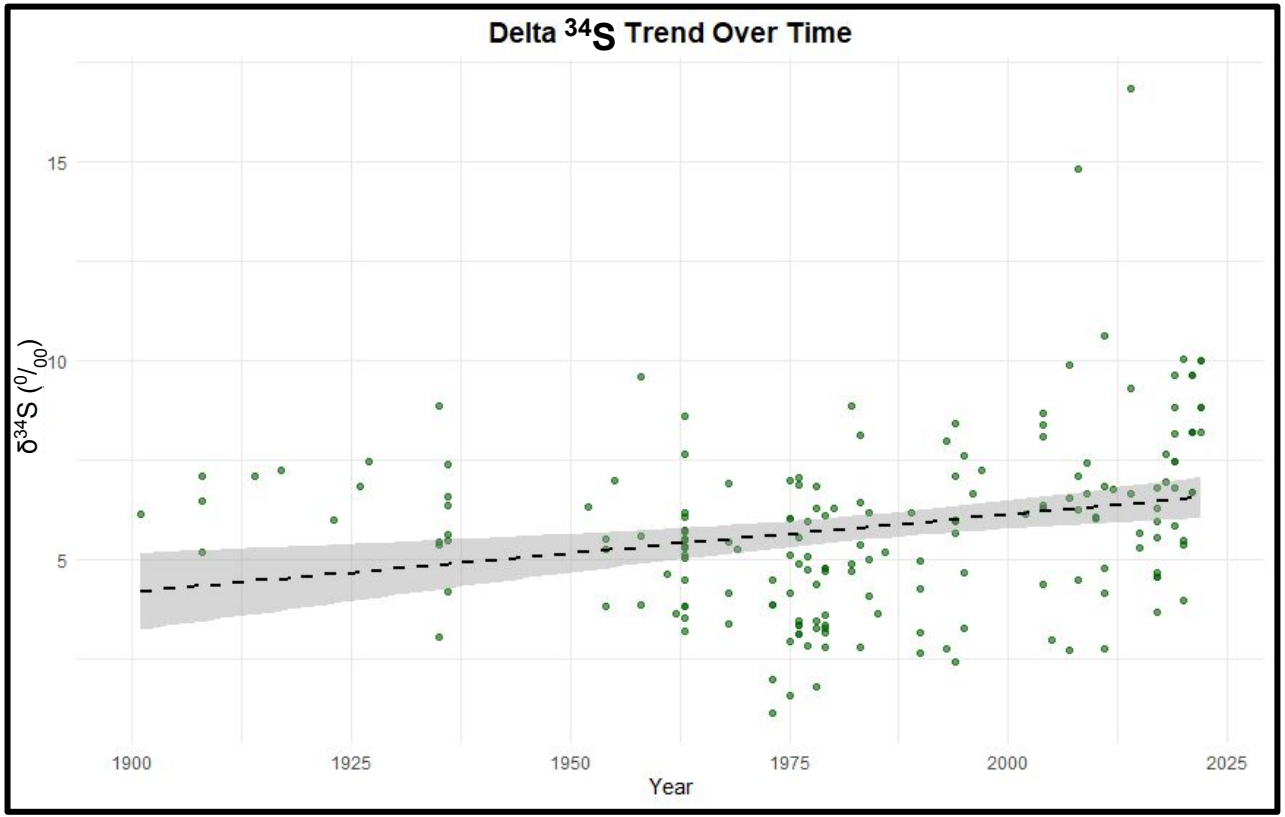
- Early sources of NO_x contamination
  - Coal (+15 NO_x)
  - **$\delta^{15}N\uparrow$**
- N sources shifted in late 20th century
  - Fossil Fuels (-9.5 NH_x, -11.5 NO_x)
  - **$\delta^{15}N\downarrow$**

Results - Sulphur



- Coal dominated power supplies in the 20th century **$\delta^{34}S$↓**
  - (Coal = **δ**+5)
- CAA implemented catalytic converters and scrubbers, causing **Less Combustion Emission**
  - **$\delta^{34}S$↑**

Results - Sulphur



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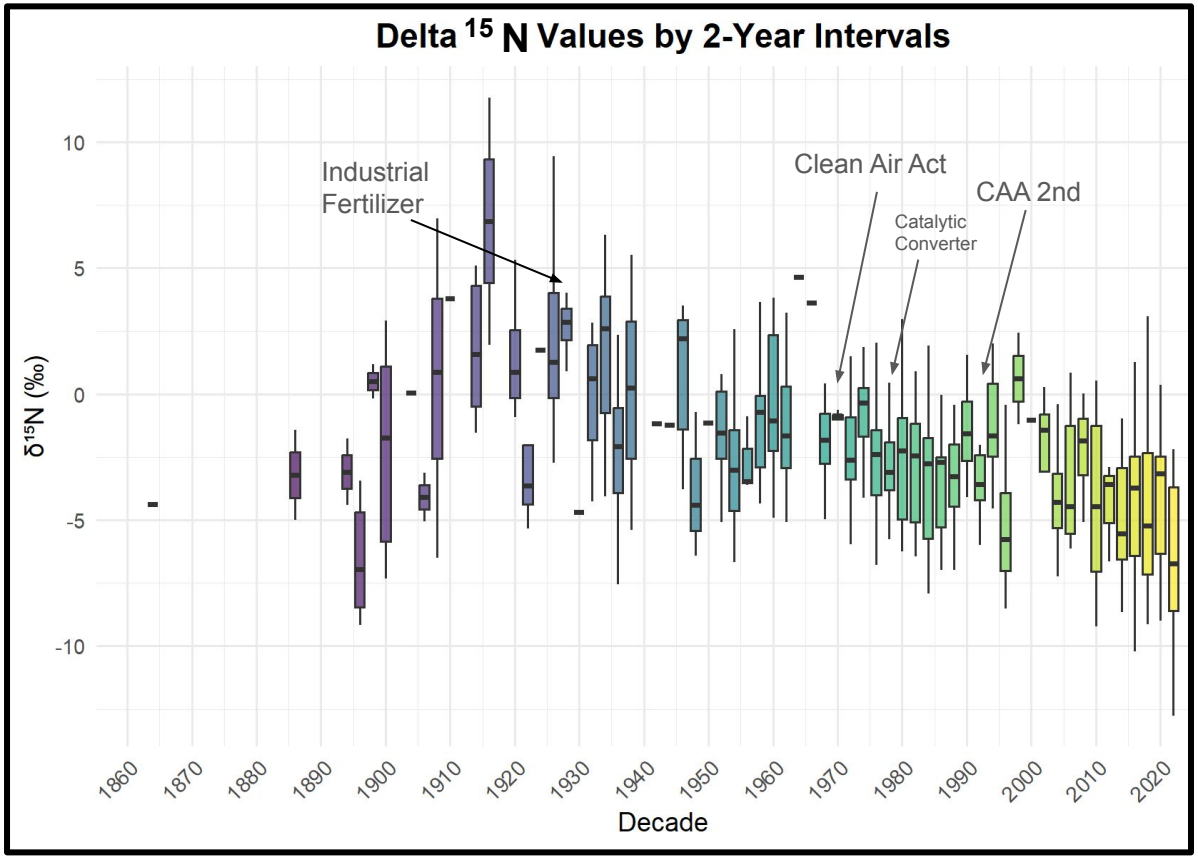
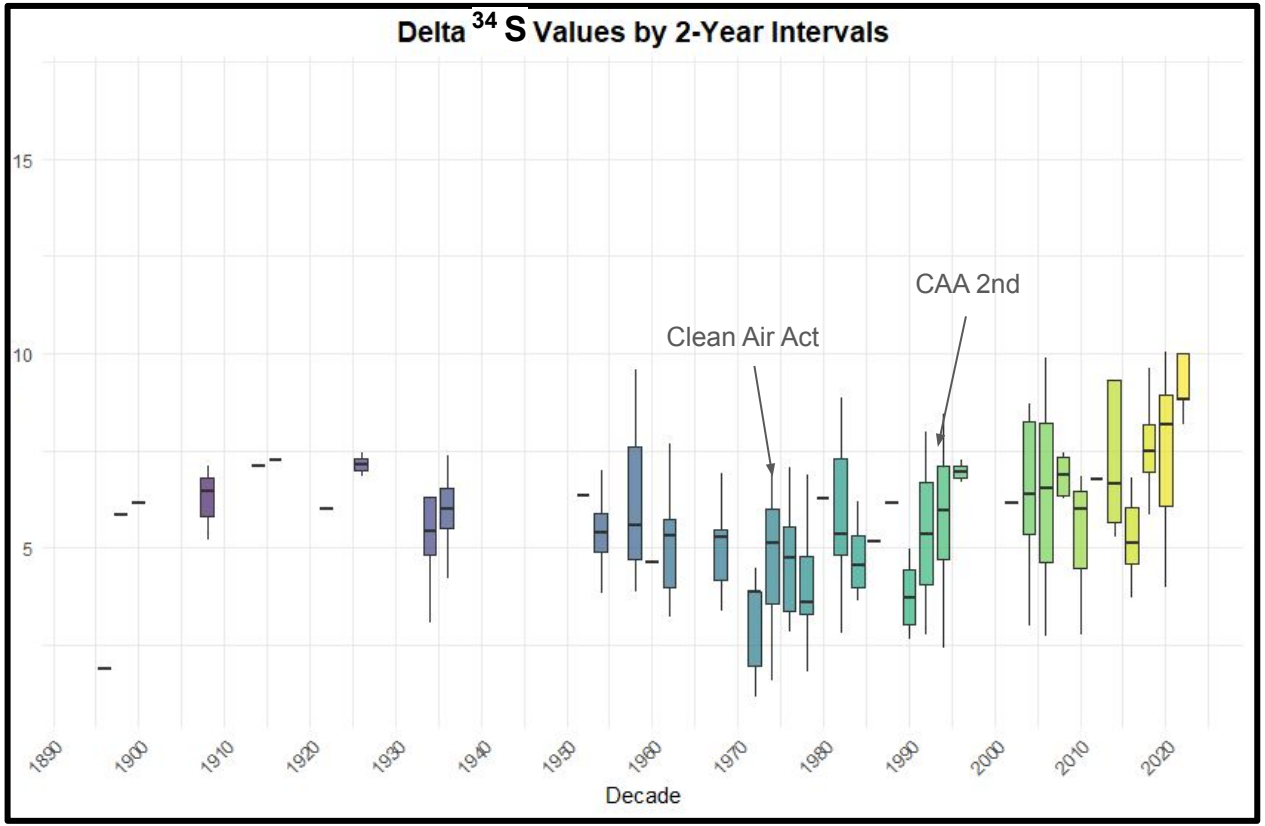
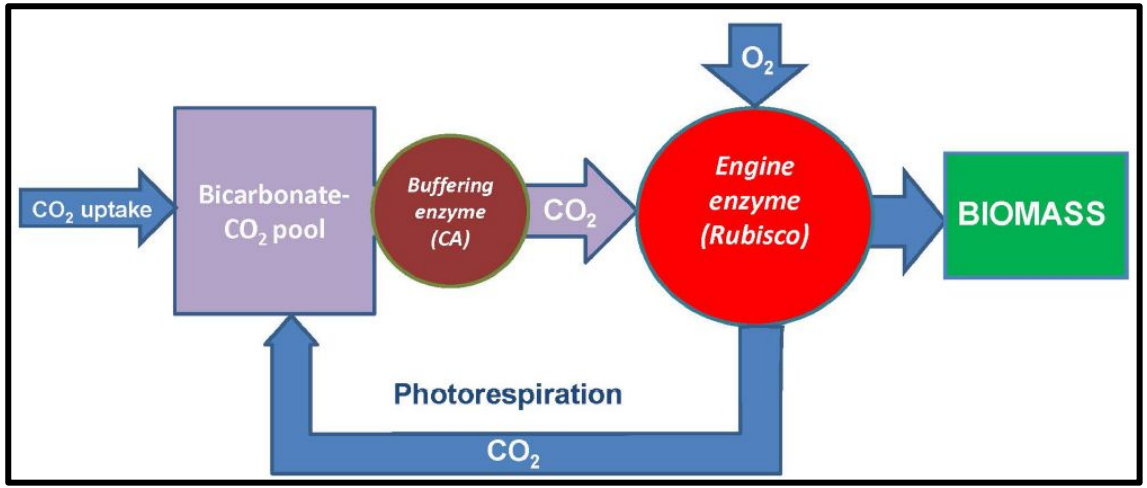
Environmental Policy Effects



Environmental Policy Effects



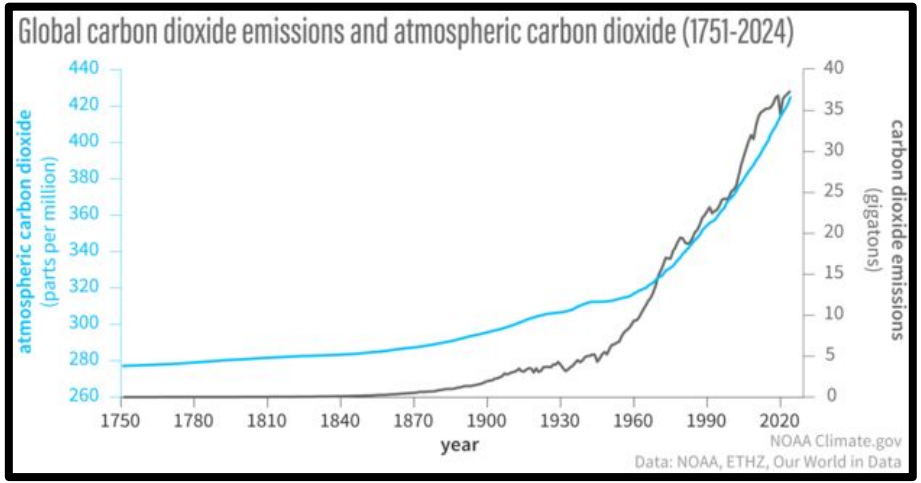
Carbon Pathways



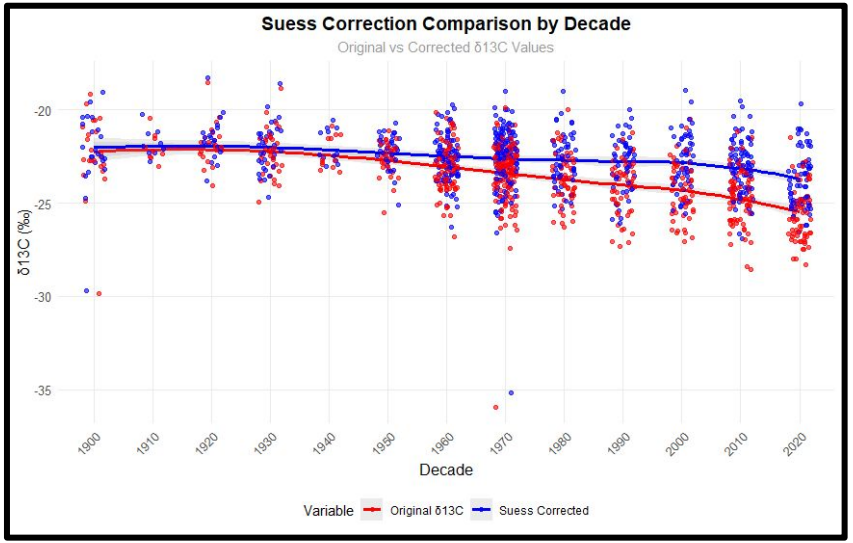
- Rubisco depletes ^{13}C during photosynthesis
- Unable to track specific source since it fractionizes within the lichen

Seuss Correcting

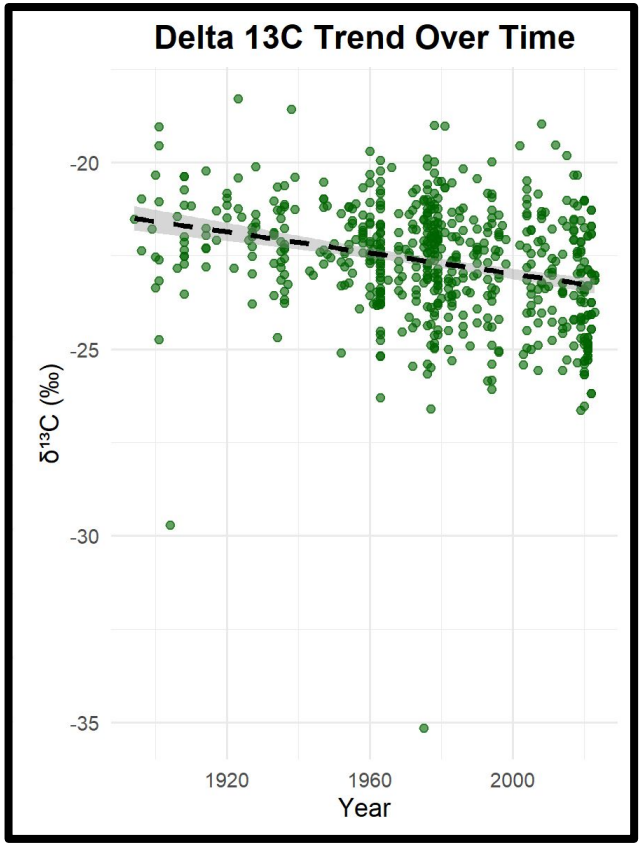
Suess correcting removes "C-depleted" CO_2 from fossil fuels as a factor from ^{13}C delta levels within lichen

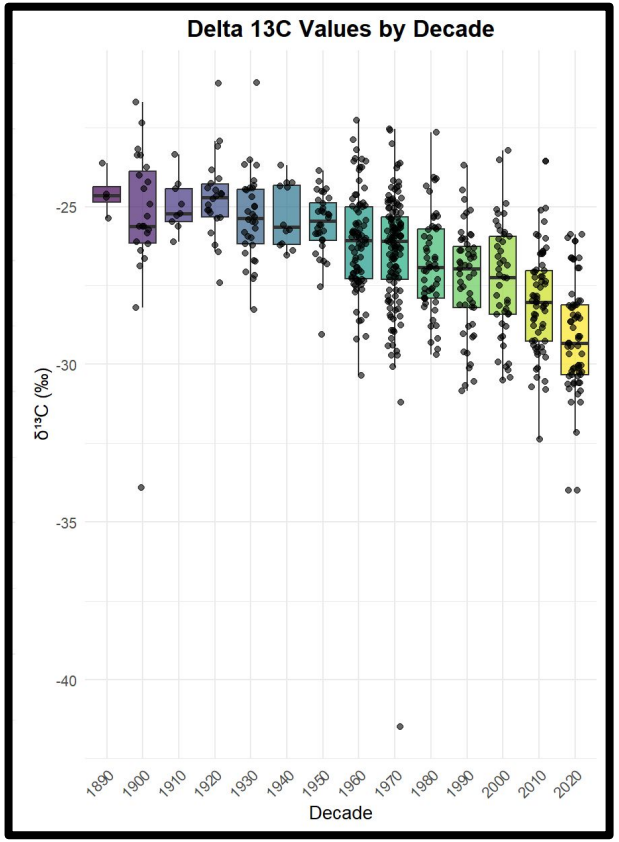


NOAA Global Monitoring Lab. (2024)



Results - Carbon

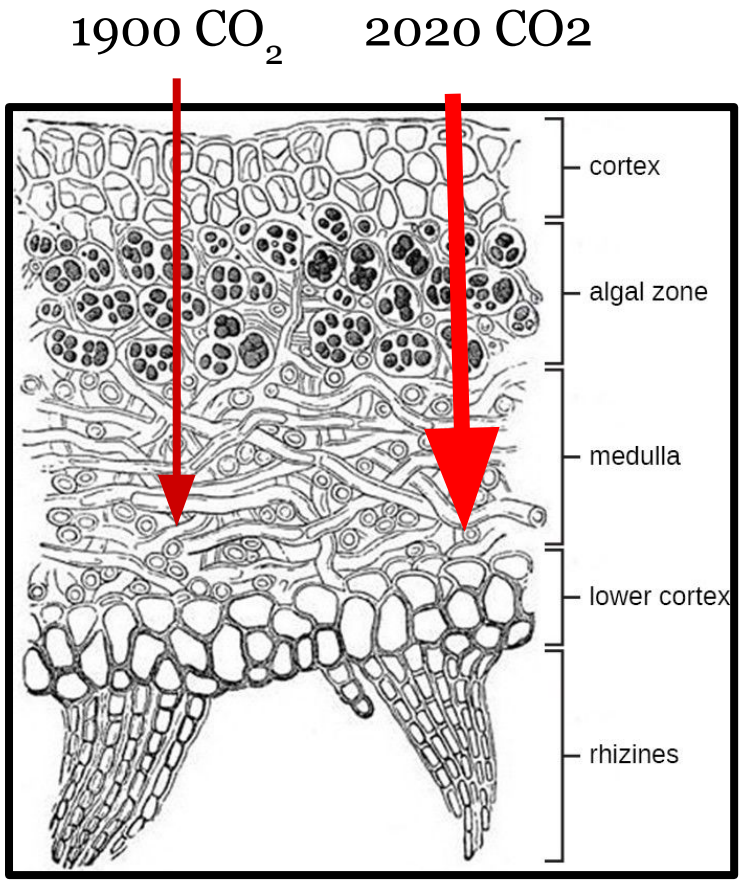




- Even after Seuss Correction, **$\delta^{13}C$**↓ in the past 20 years
- Not induced by more fossil fuels emissions, but possibly a physiological change

Physiological Change

- Lichens don't have stomata they just let gasses in through surface membranes
- Since overall fossil fuel based CO_2 **concentrations** have **increased** in the past century, lichens take up a **process larger amount into depleted ^{13}C**
- **$\delta^{13}C$↓**



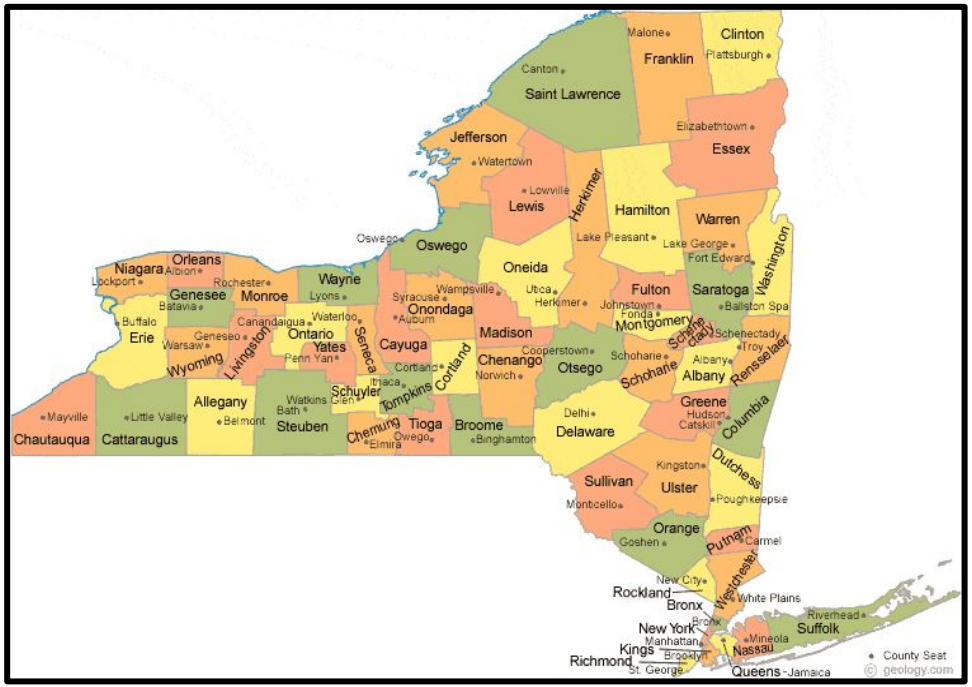
Conclusion

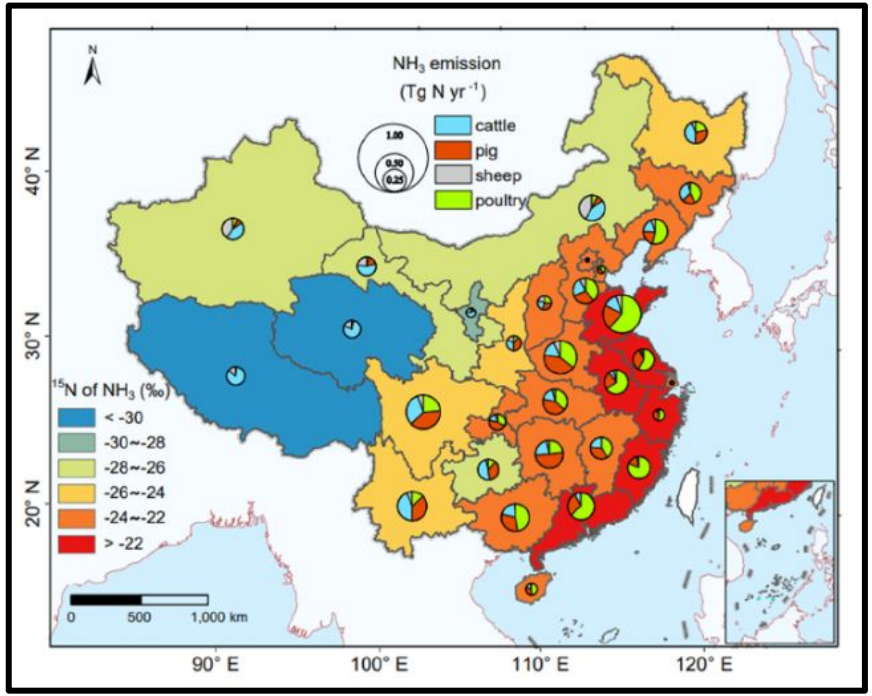
- The amount of policy towards sulphur emission in the past 30 years has proved a major difference maker in atmospheric sulphur
- CO_2 shows interesting physiological developments within lichen Carbon uptake
- Nitrogen also shows rebounding from combustion sources

Future work

- Using Monte Carlo Mixing Models to identify more accurate ranges of source contributions with Mapping

Future Work





(Liu et al. 2025)

Acknowledgements

Thank You Nexus!

I have learned a lot about what it takes to be a scientist through this program and am better of from it no matter what career pathway I take in the future. I learned how to reason with complex ideas, investigate hypothesize, and ask questions in a way to learn information in an intuitive way. I am grateful for the opportunity!

Nexus Scholars Program

SPECIAL THANKS TO THE SPARKS LAB!

Jed, Kim, Ben, Hannah,

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Seojin and Lilly,
This summer wouldn't have been the same without you guys thank you for making the work atmosphere a joy everyday!